# 1 - cd

1. $x' = \left\lfloor \frac{x-k+2p}{s} \right\rfloor + 1$, $y' = \left\lfloor \frac{y-k+2p}{s} \right\rfloor + 1$

Output size: $x' \times y' \times 1$

number of weights to learn: $k \times k \times z + 1$

For a fully-connected layer, numbers of weights: $(x \times y \times z) \times (x' \times y' \times 1)$

2.

| Convolutional layers | Fully-connected layers |
|---|---|
| Fewer parameters due to weight sharing | Very large number of parameters |
| Preserve spatial structure of images | Lose spatial information |
| Capture local patterns (edge, textures) | Mix all pixels globally |
| Translation-invariant features | No translation invarrance |

Main limit:
Convolutions only capture local relationships, ~~they~~ it's difficult for them to model long-range dependencies without deeper architectures.

3. Spatial pooling could reduce the spatial dimensions, make features more translation-invariant, reduce computational cost and overfitting, and keep only the most imporeant activations (e.g. max pooling) or average information (eg. average pooling)

4. We can reuse all convolutional and pooling layers because convolutions and pooling are position-independent operations, they can process any input size, but we can't reuse the fully-connected layers without resizing the image.

5. We can use a convolutional filter whose kernal size equals the entire input:

$$k = x, k = y$$

The convolution sees the whole input, it becomes identical to a fully-connected layer.

6. For shared convolution: $Q \times C \times 5 \times 5$

For unshared convolution: each spatial position ~~has~~ has its own weights

total parameters: $Q \times H \times W \times (C \times 5 \times 5 + 1)$



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7. • For the first convolution layer: receptive field = size of the kernel (e.g. 3×3)
   • For the second layer: it depends on the reception field of the first layer, its stride and kernel size.
   e.g. with two 3×3 convolutions, stride 1, the receptive field = 5×5.
   • Deeper layers have larger receptive fields. It means that neurons see more global features.
   • Interpretation: As depth increases, the network goes ~~to~~ from detecting edges to object parts to whole objects.

8. stride $S = 1$

$$x' = \left\lfloor \frac{x - k + xp}{s} \right\rfloor + 1 = x$$

$$\Rightarrow \quad x - k + xp - 1 = x$$

$$\Rightarrow \quad \text{the kernel size } k = \frac{6 \cdot 1}{2} = 2$$

9. $x' = \left\lfloor \frac{x - k + xp}{s} \right\rfloor + 1 = \frac{x}{2}$ and $k = 2$

We choose stride = 2, padding = 0

10.

| Layer | # Parameters | input size | output size |
|---|---|---|---|
| conv1 | $32 \times (5 \times 5 \times 3 + 1) = 2432$ | $32 \times 32 \times 3$ | $32 \times 32 \times 32$ |
| pool 1 | 0 | $32 \times 32 \times 32$ | $16 \times 16 \times 32$ |
| conv 2 | $64 \times (5 \times 5 \times 32 + 1) = 51264$ | $16 \times 16 \times 32$ | $16 \times 16 \times 64$ |
| pool 2 | 0 | $16 \times 16 \times 64$ | $8 \times 8 \times 64$ |
| conv 3 | $64 \times (5 \times 5 \times 64 + 1) = 102464$ | $8 \times 8 \times 64$ | $8 \times 8 \times 64$ |
| pool 3 | 0 | $8 \times 8 \times 64$ | $4 \times 4 \times 64$ |
| fc 4 | $(4 \times 4 \times 64 + 1) \times 1000 = 1025000$ | $4 \times 4 \times 64$ | 1000 |
| fc 5 | $(1000 + 1) \times 10 = 10010$ | 1000 | 10 |

Total parameters

$= 2432 + 51264 + 102464 + 1025000 + 10010$

Most parameters come from fully connected layers

While convolution layers dominate computational cost but not parameter count.

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11. Total parameters

$= 2432 + 51264 + 102464 + 102500 + 10010 \approx 1.19$ million parameters

CIFAR-10 training samples $= 50000$

The model has 24× more parameters than training samples.

Therefore, the model has high capacity and it can model complex patterns, and relationships, but it risks of overfitting, so we should add regularization, dropout or data augmentation.

12. SVM: # params $= 10 \times (32 \times 32 \times 3 \times 10 + 1) = 30730$

It only learns a seperating hyperplan, there is no hierarchy

BoW + SVM: # params $= 10 \times (K + 1) = 10K + 10$

If we also count the codebook as learned parameters, and the local descriptor has dimension D, then, codebook: $K \times D$

Total: $K \cdot D + 10K + 10$

14. Training computes gradients and update parameters, while in the test mode, the code does not do backpropagation, it only tests performance

16. learning rate: • too small → slow convergence
• too large → unstable training or divergence

It controls step size of wight update

batch size: • too small → noisy gradient, slower per epoch
• too large → smoother gradient, but less generalization

It controls variance of gradient estimation

17. At the beginning, the accuracy is low and the loss is high because the model predictions are random and each image has 10 possible classes.

Interpretation: The model starts untrained and learns progressively during training.

18. The training accuracy is very hight, but the test accuracy is low.
This phenomenon is overfitting.

Interpretation: the network memorizes the training data but fails to generalize.
This is typical when the model is complex or the dataset is small.



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19. The model overfits significantly after epoch 7. Training accuracy reaches 100%, but test accuracy is around 75%. The test loss stops decreasing and starts to rise. This phenomenon confirms that the model is memorizing the training data instead of generalizing. But its overfitting degree is better than not using the method.

20. To prevent data leakage. The validation set must remain unseen. Using its statistics (mean, variance) for normalization would "leak" information to the model and give an unrealistically high performance score.

21. • Standardization: It is simple and normalizes each color channel independantly. It igonores correlations between channels.
- ZCA : It is complex, It normalizes and decorrelates all features. It removes statistical redudancy.
- Comparison: ZCA is computationally expensive but can help learning by removing redundacy. Standardization is simpler and more efficace.

22. Data augmentation significantly improved the model and solved the overfitting problem. Previously, training accuracy was very high while test accuracy stalled at 75%, and the test loss increased. Now, the training and test curves are now very close, Test loss decreases along with training loss. The model now generalizes much better.

23. No, it's not usable for all images.
It works well for natural images like in CIFAR-10 where a flipped object is still the same object.
It must not be used when the horizontal direction has meaning. For example, we can't apply this model to images of text or numbers.

24. The main limits is that these transformations don't create new, original information. The model only sees variations of the same images. If the original dataset is small or biased, augmentation can't solve the problem of overfitting.



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25. Other methods possible:

- ColorJitter: It's effective. It prevents overfitting well.
- RandomRotation: It's effective. The training is stable.
- Random Erasing: It's a bit effective.

26. The result is a much smoother test loss curve and a higher accuracy. It's much more stable. Previously, the test loss curve would be bumpy and struggle to find a good minimum

27. A high learning rate at the start allows for fast convergence
A low learning rate at the end allows for fine-tuning to settle into a better, more stable minimum without overshooting it.

28. Other Optimizers (SGD Variants):

- Adam
- RMSprop

Other Schedulers (learning rate planning)

- StepLR
- ReduceLROnPlateau
- CosineAnnealingLR

29. Compared to the previous model, training accuracy is a bit lower and training loss is higher. This is because dropout makes training harder. The gap between train and test curves becomes smaller and the curves become smoother.

30. Regularization discourages a model from becoming too complex or just memorizing the training data. It helps the model generalize better to new, unseen data and prevent overfitting.

31. 1) Ensemble Averaging: Dropout is training thousands of different, small "sub networks" at the same time since different neurons are dropped at each pass). The final result is an average of all their predictions, which is more robust

2) Prevent Co-adaptation: It stops neurons from lying too much to on specific other neurons. It forces each neuron to learn more robust features that are useful on their own.



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32. The hypermeter is p (dropout rate).

A low p provides week regularization.

A high p provides very strong regularization. If p is too high, it can slow down training too much or even cause underfitting.

33. During Training: Dropout is active. It randomly sets a fraction p of its inputs to zero and scales up the remaining inputs by $\frac{1}{1-p}$.

During Testing: Dropout is inactive. It does nothing and just passes all the data through without dropping anything.

34. Batch normalization dramatically speeds up training. The model reach the same accuracy in far fewer epochs compared to the previous models.

The training is much more stable. ~~Batch normalization makes the model far less sensitive to the choice of learning rate,~~ and the test loss curve is much smoother.

Batch normalization also acts as a regularizer, so the final test accuracy is higher and it helps prevent overfitting and improves generalization.

## Summary of Implemented Methods

(1) Stardardization

- Principal: rescales the input data by subtracting the mean and dividing by the standard deviation of the training set.
- Why it helps: it centers the data around zero, which helps the optimizer converge faster and makes the training more stable.

(2) Data Augmentation (Random Crop & Flip)

- Principal: creates new training examples by applying random transformations, like cropping and horizontal flipping to the original images
- Why it helps: it increases the size and diversity of the training set, so it reduces overfitting.

(3) Advanced Data Augmentation ( Color Jitter, Rotation, Erasing)

- Principle: applies more complex transformations like randomly changing brightness or erasing parts of the image.



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- Why it helps: it forces model to learn more robust features

(4) Dropout

- Principal: randomly deactivates a fraction of neurons during each training pass.
- Why it helps: prevents neurons from relying too much ~~to~~ on each other and forces the network to learn more independant, robust features.

(5) Learning Rate Scheduler

- Principal: automatically reduces the learning rate during training
- Why it helps: it allows for fast learning with a high learning rate at the start and stable fine-tuning with a low learning rate at the end.

(6) Adam Optimizer

- Principal: an advanced optimizer that adapts the learning rate for each parameter individually based on past gradient information.
- Why it helps: it converges much faster and is less sensitive to the initial learning rate choice than standard SGD

(7) Batch Normalization

- Principal: normalizes the output of each layer within the network by using the mean and standard deviation of the current batch.
- Why it helps: it stabilizes the learning process, ~~allows for much higher learning rates and~~ significantly accelerates training convergence.



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